



DOMAIN TRACK

Gen AI & Cloud Computing

TEAM COMPOSITION

Team of 6 (Vibe Coding)

LIVE DEPLOYMENT

klyvix-177846015206.europe-west1.run.app

SWAGGER DOCS

/docs (Interactive Open API)

6

SPECIALIZED PERSONAS

<1.5s

TIME TO FIRST TOKEN

0%

DOC HALLUCINATION

99.9%

MULTI-LLM UPTIME

1 Project Problem & Value Proposition

Standard consumer chatbots suffer from *single-model fatigue*: they dump generic text when teaching, produce unstructured code feedback, hallucinate on uploaded documents, and crash when API rate limits occur. **Klyvix** replaces generic chat with an enterprise-grade AI Command Center powered by 6 specialized personas, dense vector retrieval via Qdrant, and an automated multi-LLM failover engine.

2 The 6 Specialized AI Workspaces

Study Mentor

Socratic/Direct

Step-by-step guided tutor with LaTeX math typesetting, verification check-ins, and direct exam mode.

Code Reviewer

JSON Schema

Systematic multi-dimensional audit covering Bugs, Security, Performance, and Style with fix snippets.

Code Colleague

Pair Eng.

Senior engineering partner writing production code, surgical diffs, and architectural trade-off evaluations.

Document Analyzer

Qdrant RAG

Dense vector retrieval over uploaded PDFs (768-dim embeddings). Strict document-bound grounding with zero hallucination.

Resume Reviewer

ATS Audit

Actionable career critique with Strengths, Gaps, and concrete Before/After quantified rewrites.

Research Assistant

Search Grounded

Multi-source literature synthesis powered by Google Search grounding with consensus and conflict detection.

3 Technology Stack & Architecture Highlights

Domain	Selected Tech	Key Production Capability
Backend & Streaming	FastAPI (Python 3.12) + Uvicorn + SSE	Asynchronous Server-Sent Events (SSE) for zero-latency token streaming to the browser.
Multi-LLM Routing	Gemini 2.0 Flash + Mistral + Groq Llama	Dynamic routing with automatic sub-second failover on rate limits (429) or network timeouts.
Vector Database (RAG)	Qdrant Cloud + 768d Gemini Vectors	Semantic chunking and cosine similarity search for strict grounded document Q&A.
Cloud Hosting	Google Cloud Run (Docker Container)	Serverless autoscaling (0-10 instances), auto-HTTPS, TLS 1.3 at europe-west1.

4 Evaluator & Submission Verification Links

Production Application

Live URL: <https://klyvix-177846015206.europe-west1.run.app/>
Instant web access with zero setup, dark/light theme toggle, and live persona chats.

OpenAPI / Swagger Docs

Swagger UI: <https://klyvix-177846015206.europe-west1.run.app/docs>
Interactive REST API documentation for backend testing and endpoint validation.